

TP_MODUL_04

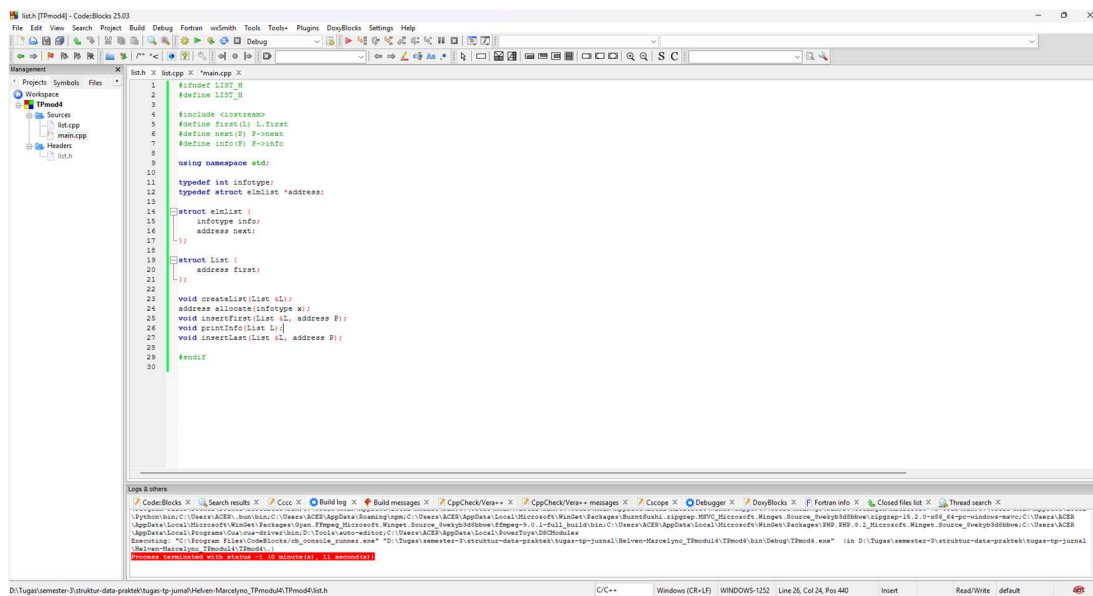
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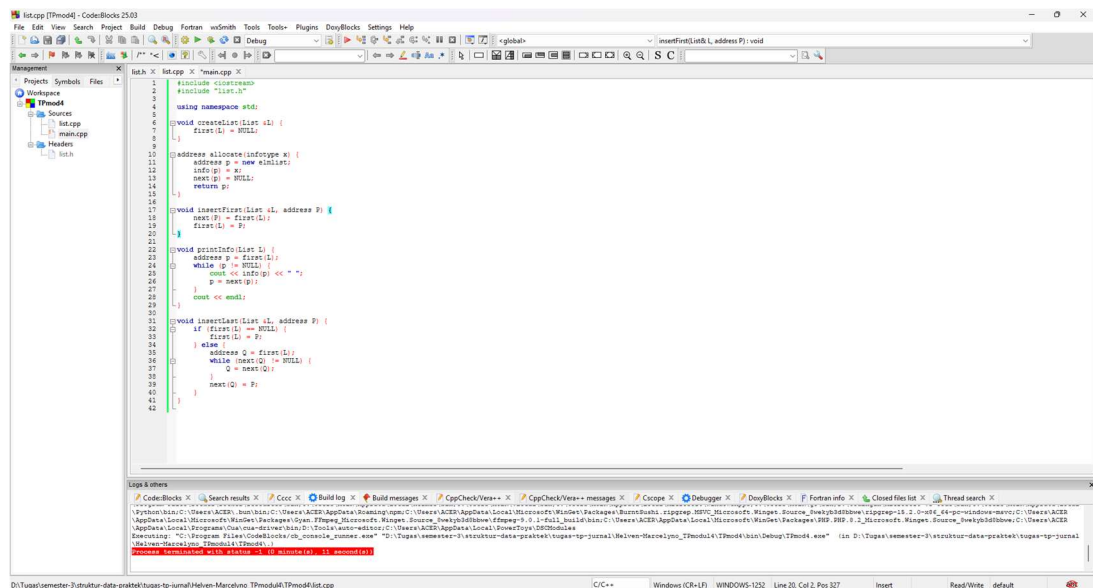
FILE CODEBLOCKS

1. List.h



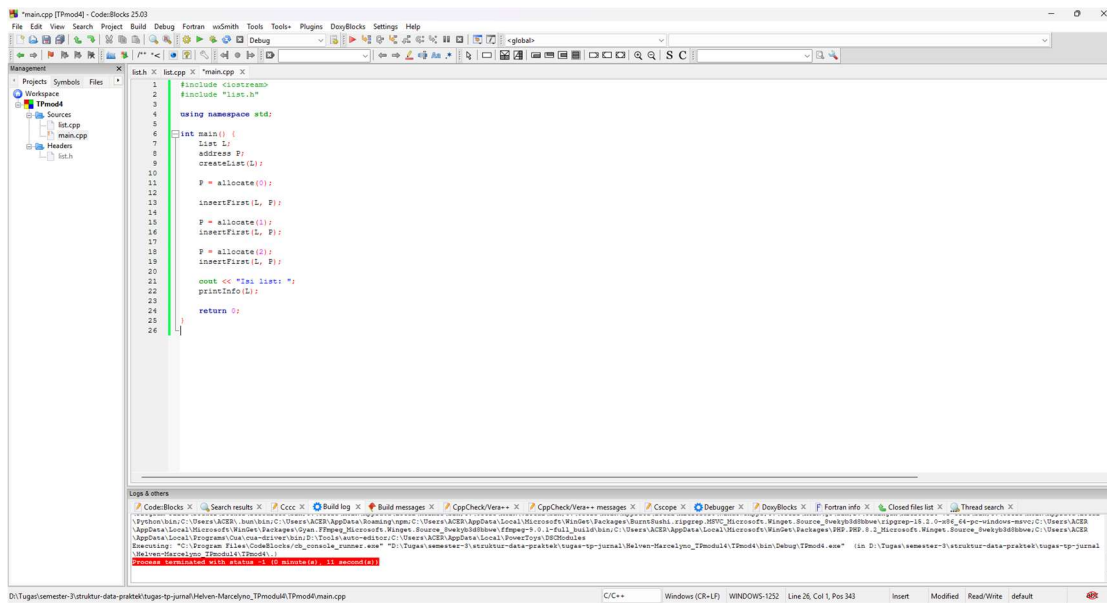
```
1 #ifndef LIST_H
2 #define LIST_H
3
4 #include <iostream>
5 #define first(i) l.first
6 #define next(i) i->next
7 #define info(i) i->info
8
9 using namespace std;
10
11 typedef int infotype;
12 typedef struct elmlist *address;
13
14 struct elmlist {
15     infotype info;
16     address next;
17 }
18
19 struct List {
20     address first;
21 }
22
23 void createList(List &l);
24 address allocate(infotype x);
25 void insertFirst(List &l, address p);
26 void printInfo(List &l);
27 void insertLast(List &l, address p);
28
29 #endif
```

2. List.cpp



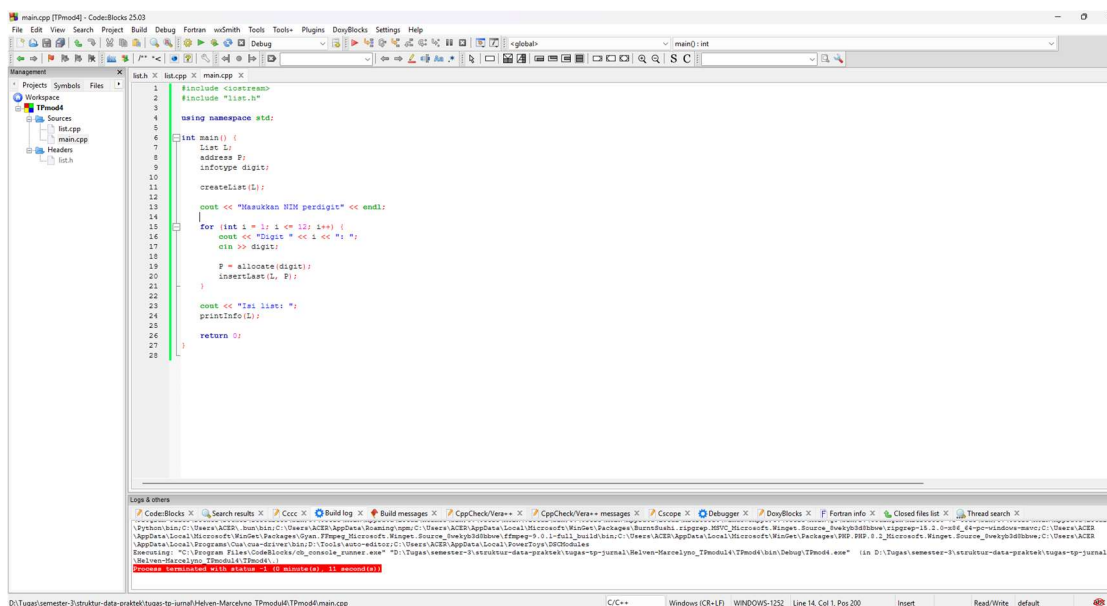
```
1 #include <iostream>
2 #include "List.h"
3
4 using namespace std;
5
6 void createList(List &l) {
7     first(l) = NULL;
8 }
9
10 address allocate(infotype x) {
11     address p = new elmlist;
12     info(p) = x;
13     next(p) = NULL;
14     return p;
15 }
16
17 void insertFirst(List &l, address p) {
18     next(p) = first(l);
19     first(l) = p;
20 }
21
22 void printInfo(List &l) {
23     address p = first(l);
24     while (p != NULL) {
25         cout << info(p) << " ";
26         p = next(p);
27     }
28     cout << endl;
29 }
30
31 void insertLast(List &l, address p) {
32     if (first(l) == NULL) {
33         first(l) = p;
34     } else {
35         address q = first(l);
36         while (next(q) != NULL) {
37             q = next(q);
38         }
39         next(q) = p;
40     }
41 }
42
43 }
```

3. main.cpp 3 digit nim terakhir



```
1 #include <iostream>
2 #include "list.h"
3
4 using namespace std;
5
6 int main() {
7     List l;
8     address p;
9     createList(l);
10
11     p = allocate(0);
12     insertFirst(l, p);
13
14     p = allocate(1);
15     insertFirst(l, p);
16
17     p = allocate(2);
18     insertFirst(l, p);
19
20     cout << "Tas List: ";
21     printInfo(l);
22     return 0;
23 }
```

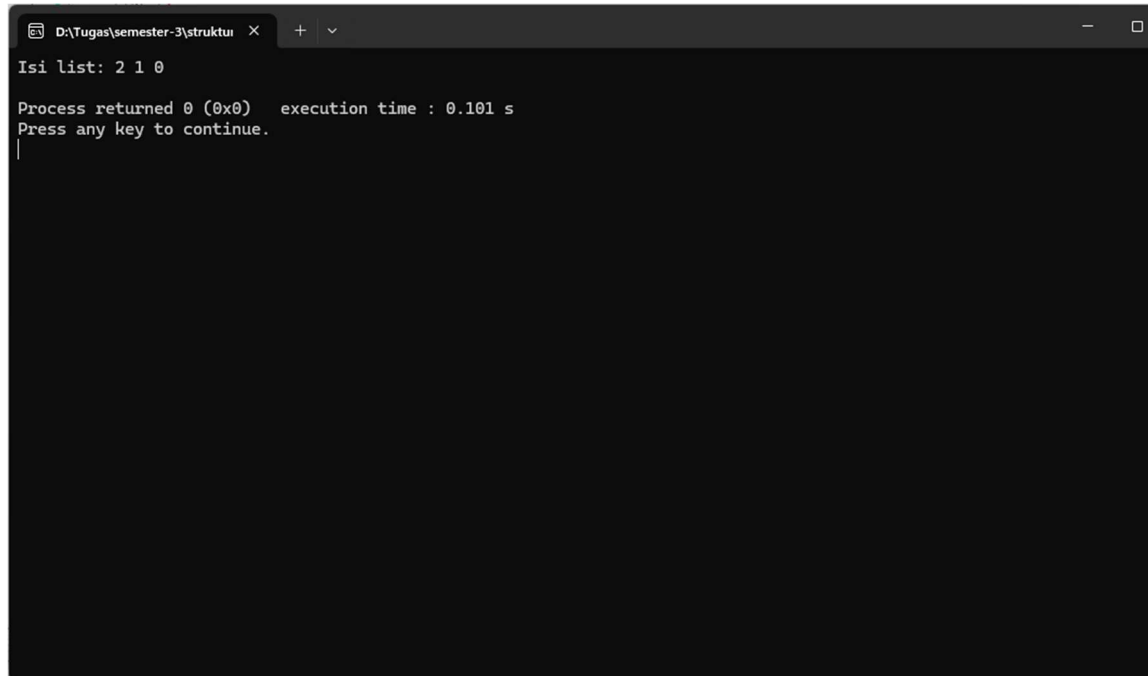
4. Main.cpp sesi have fun



```
1 #include <iostream>
2 #include "list.h"
3
4 using namespace std;
5
6 int main() {
7     List l;
8     address p;
9     int digit;
10     createList(l);
11
12     cout << "Masukkan NIM perdigit" << endl;
13
14     for (int i = 1; i <= 10; i++) {
15         cout << "Digit " << i << ": ";
16         cin >> digit;
17         p = allocate(digit);
18         insertLast(l, p);
19     }
20
21     cout << "Tas List: ";
22     printInfo(l);
23     return 0;
24 }
```

HASIL CODE & RUN

1. HASIL 3 DIGIT TERAKHIR NIM

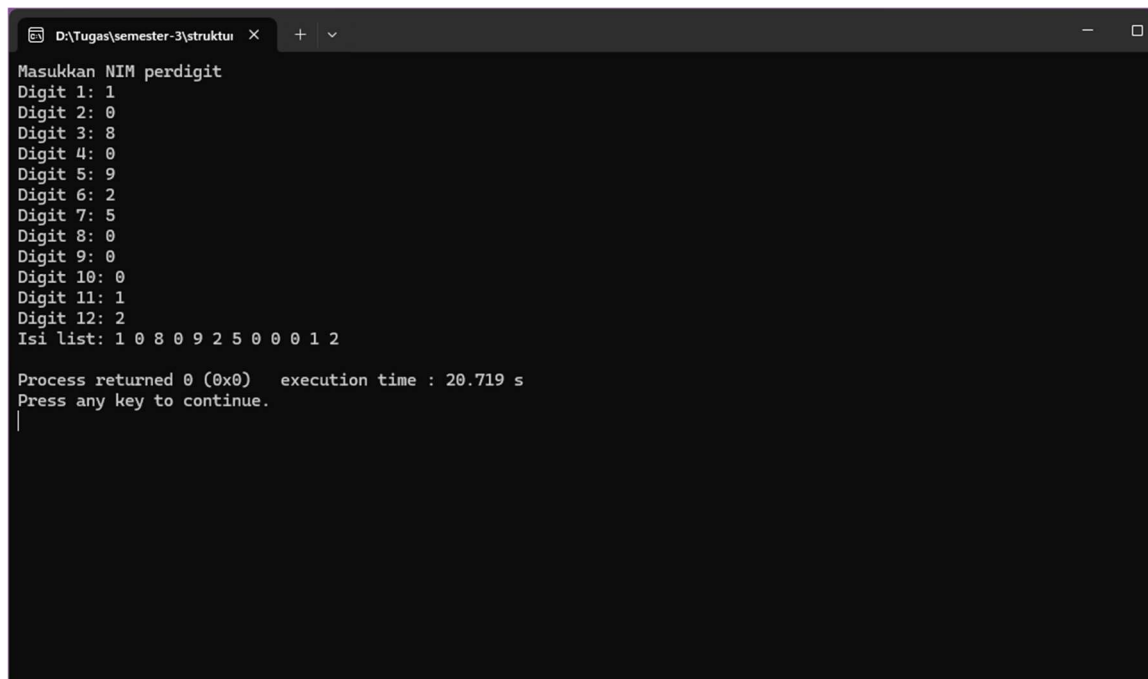


```
D:\Tugas\semester-3\struktur x + v
Isi list: 2 1 0

Process returned 0 (0x0)   execution time : 0.101 s
Press any key to continue.
|
```

The screenshot shows a Windows command prompt window with a dark background. The title bar indicates the file path is D:\Tugas\semester-3\struktur. The program has executed and printed 'Isi list: 2 1 0'. Below this, it shows the process return code as 0 (0x0) and the execution time as 0.101 seconds. The prompt 'Press any key to continue.' is displayed, followed by a vertical bar character '|' on the next line.

2. HASIL SESI HAVE FUN



```
D:\Tugas\semester-3\struktur x + v
Masukkan NIM perdigit
Digit 1: 1
Digit 2: 0
Digit 3: 8
Digit 4: 0
Digit 5: 9
Digit 6: 2
Digit 7: 5
Digit 8: 0
Digit 9: 0
Digit 10: 0
Digit 11: 1
Digit 12: 2
Isi list: 1 0 8 0 9 2 5 0 0 0 1 2

Process returned 0 (0x0)   execution time : 20.719 s
Press any key to continue.
|
```

The screenshot shows a Windows command prompt window with a dark background. The title bar indicates the file path is D:\Tugas\semester-3\struktur. The program prompts the user to 'Masukkan NIM perdigit'. It then displays 12 digits, one per line, from 'Digit 1: 1' to 'Digit 12: 2'. Below these, it prints 'Isi list: 1 0 8 0 9 2 5 0 0 0 1 2'. The process return code is 0 (0x0) and the execution time is 20.719 seconds. The prompt 'Press any key to continue.' is shown, followed by a vertical bar character '|' on the next line.